Desymmetrization of Quantum Spectra: A Bayesian Sector-Count Diagnostic with Measured Coverage

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Abstract

Spacing statistics are the standard diagnostic for quantum chaos, but they are measured on a superposition whenever the Hamiltonian commutes with an involution. A spectrum that appears Poisson-like may instead be a union of unresolved sectors, and the standard mean gap ratio $\langle r \rangle$ cannot distinguish the two. We present a diagnostic that verifies the symmetries, splits the spectrum into joint sectors, flags a contaminated full spectrum, and estimates the unresolved-sector count as a posterior with measured coverage. The mean cannot resolve the count, because the per-spectrum scatter exceeds the gaps between counts; the sequence of gap ratios carries more. A first-order Markov model fitted on simulated superpositions yields a posterior over `{1, ..., 6, ≥6/poisson}`. At a nominal 95% interval, measured coverage is `0.99`, `0.92`, `0.91`, `0.96`, `0.98` at `1, 2, 3, 4, 6` sectors and `0.78` for a Poisson spectrum, over 400 spectra per row. The single-sector case is decisive (`p = 0.97`, interval `1..1`); the two-sector case yields a lower bound (2) but not a count, and from three sectors upward the count is not recoverable from adjacent-gap ratios alone, because `3` through `6` are indistinguishable to them. Two-sector GOE and GUE are indistinguishable in the full spectrum, and that limit is measured rather than assumed: the best single-spectrum class accuracy is `0.79` at one sector and `0.53`, which is chance, at two, while averaging thousands of spectra sharpens the mean to `0.0006` and leaves the single measurement untouched. The false-positive rate is stated (`37` of `400` pure single-sector GOE spectra are flagged, `0` of `400` GUE), and the limits are reported rather than hidden. The companion dynamical analysis shows that the double-bracket flow preserves sector populations only for a state that commutes with the involution, so the flow cannot repair a contaminated spectrum.

1 Introduction

1.1 Problem statement

Quantum chaos is diagnosed by level-spacing statistics [4]. The mean adjacent-gap ratio $\langle r \rangle$ [6, 5] takes three reference values:

class	$\langle r \rangle$	meaning
Poisson	`0.386`	integrable
GOE	`0.531`	chaotic, time-reversal symmetric
GUE	`0.600`	chaotic, time-reversal broken

When a Hamiltonian commutes with an involution σ, its spectrum is the union of the spectra of the σ `= ±1` sectors. If the sectors are not resolved, $\langle r \rangle$ of the full spectrum is not the class of the Hamiltonian. It is the class of a superposition.

The failure is silent. There is no error message. The number `0.42` looks like a legitimate measurement. It is wrong, and it is wrong in a specific way: it is exactly what a two-sector superposition produces.

1.2 Origin of the diagnostic

This diagnostic was not designed in the abstract and justified afterwards. It is the product of a larger project, and the failure that produced it is the failure it detects.

The project began with a candidate operator for the Riemann hypothesis. The map σ`(s) = 1 − conj(s)` reflects the complex plane about the critical line; its fixed locus is exactly `Re(s) = 1/2`; and a Hamiltonian built around this involution was proposed as a route to the zeros. That proposal, the dimension-shift hypothesis, was tested against criteria it had set for itself and was abandoned. The decisive run found that the sector class of a σ-symmetric Hamiltonian is inherited from the input blocks, not produced by the σ structure: GOE sectors at `0.528 ... 0.533`, GUE sectors at `0.596 ... 0.602`, but a full spectrum at `0.420 ... 0.425` for either input. The verdict is recorded with the abandoned hypothesis and is final.

Two errors made during that work are the subject of this paper. The first was statistical: a single seed of a 90-point grid reported "supported", and 300 seeds with error bars showed that the value was noise (`0.523 ± 0.069`). The second was structural, and it is the one that matters here: the model's universality class was tested on the full spectrum of a σ-symmetric Hamiltonian, and a GOE model appeared to reach GUE statistics. The apparent GUE class was not a property of the model. A Hamiltonian that commutes with an involution has a spectrum that is the union of its sector spectra, and that union has a mean gap ratio of about `0.42` for either input class. The class was being read from a superposition.

Neither error was a bug in one codebase. Both are ways of reading a property from a sample that does not carry it. This paper makes the structural one visible, measures when the information is and is not recoverable, and states the limits.

1.3 Measured dependence on the number of unresolved sectors

At 4000 seeds per point, `60` levels per sector, the full-spectrum $\langle r \rangle$ depends on the number of unresolved sectors (Fig. 1):

unresolved sectors	GOE blocks	GUE blocks
1	`0.530`	`0.599`
2	`0.423`	`0.422`
3	`0.404`	`0.400`
4	`0.396`	`0.393`
6	`0.391`	`0.389`

Each entry is the mean over `4000` simulated spectra and carries a standard error of at most `0.0007`, which is three decimal places of real information. The per-spectrum scatter at the same points is `0.017` to `0.039` (Appendix A), that is twenty-five to sixty times the standard error of the mean. The whole argument of Section 1.4 rests on that ratio.

Two facts follow:

- **At three or more sectors, $\langle r \rangle$ is inside the Poisson window** `[0.36, 0.41]`. A researcher who measures `0.39` and concludes "integrable" is wrong, and the standard test gives no indication.
- **At two sectors the class is erased.** `0.4232` against `0.4222` is a gap of `0.0010`, which is `1.4` standard errors of the two means, and one thirtieth of the per-spectrum scatter of `0.031`. A class test on a two-sector full spectrum is uninformative, not merely noisy: averaging more spectra cannot rescue it, and Section 1.4 measures why.

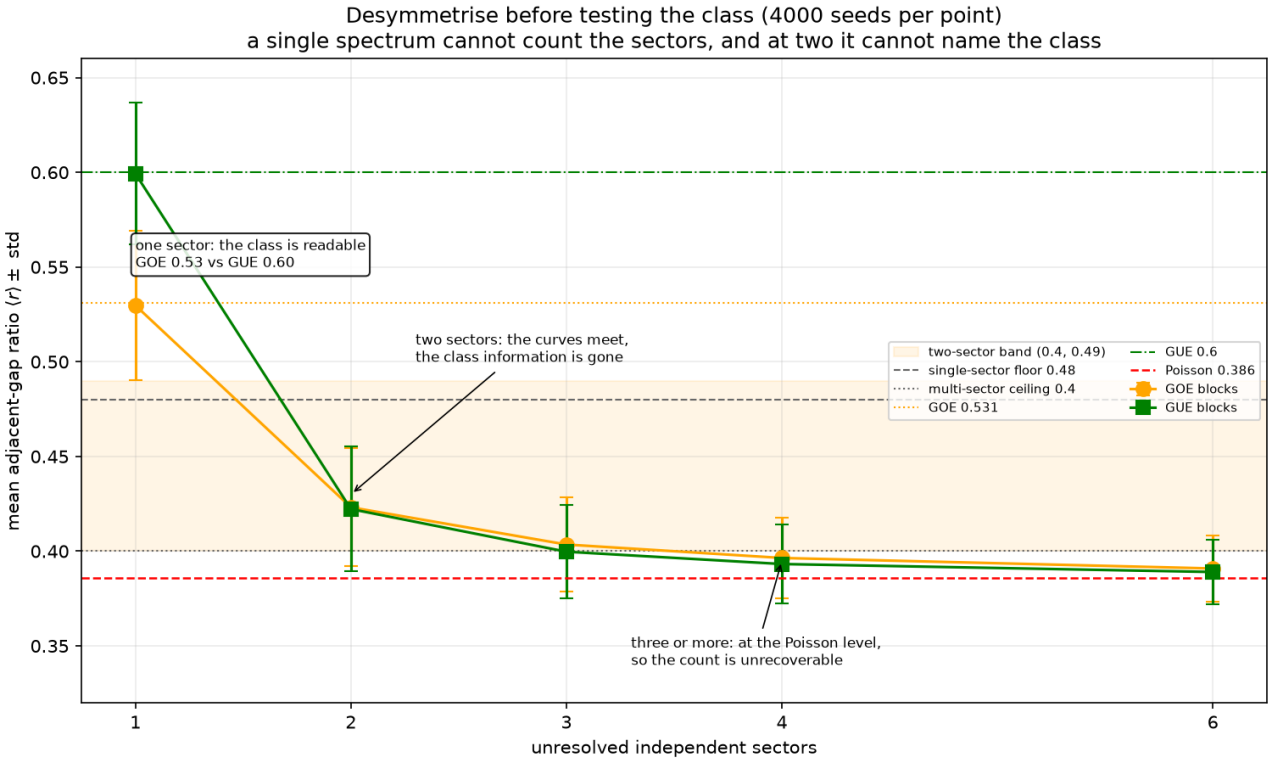


Figure 1: Full-spectrum mean adjacent-gap ratio against the number of unresolved independent sectors, at 4000 seeds per point and 60 levels per sector. GOE input blocks (circles) and GUE input blocks (squares). Reference values are marked: Poisson 0.386, GOE 0.531, GUE 0.600. The shaded band is the Poisson window [0.36, 0.41]; from three sectors upward the full spectrum lies inside it.

1.4 Statistical resolution of the diagnostic

The table above is an average over thousands of spectra, and a measurement is one spectrum. The two are different questions, and the diagnostic is only useful for the second, so the resolution of the statistic is measured rather than assumed (Fig. 2).

What more spectra buy. Fig. 2(a) repeats the two-sector measurement on nested samples of 25 to 3200 spectra per class:

spectra per class	GOE $\langle r \rangle$	GOE 95% error	GUE $\langle r \rangle$	GUE 95% error
25	0.4332	0.0119	0.4145	0.0144
100	0.4284	0.0061	0.4139	0.0067
400	0.4253	0.0030	0.4208	0.0032
1600	0.4241	0.0015	0.4227	0.0016
3200	0.4231	0.0011	0.4227	0.0011

Three things are visible in it. The mean **converges**: the GUE value moves from 0.4145 at 25 spectra to 0.4227 at 3200, so 25 spectra are off by 0.008 and 100 spectra still carry a 95% error of 0.007. The 95% error **shrinks as the sample grows**: 128 times the spectra (25 to 3200) reduces the GUE error by a factor of 12.6 and the GOE error by 11.0, either side of the 11.3 expected from the inverse square root of the sample size. The per-spectrum scatter **does not move**: 0.0305 to 0.0330 for GUE at every sample size, because it is a property of one spectrum and not of the sample.

What that means for one spectrum. Fig. 2(b) superimposes the two per-spectrum distributions at two sectors, 4000 spectra per class, and measures the best a single threshold on $\langle r \rangle$ can do. The answer is 0.526 balanced accuracy against the chance value 0.500: the two-sector classes are interchangeable one spectrum at a time. The same measurement at one sector, where the classes are 0.530

and 0.599, gives 0.790.

The two results are consistent and they are not in tension. In the 4000-spectrum table of the previous section the two-sector class gap is 0.0010, which is 1.4 standard errors; in the nested series of Fig. 2 at 3200 spectra it is 0.0004, which is 0.5 standard errors. Neither is a measurement of a class difference: both are consistent with zero, and both are one thirtieth of the scatter that one spectrum carries. **More spectra sharpen the mean and leave the single measurement untouched.** That is why the diagnostic resolves the symmetry structurally, by splitting the spectrum into sectors, instead of averaging harder statistically: the averaging route is bounded by the overlap of the distributions, not by the number of spectra, and this paper measures the bound.

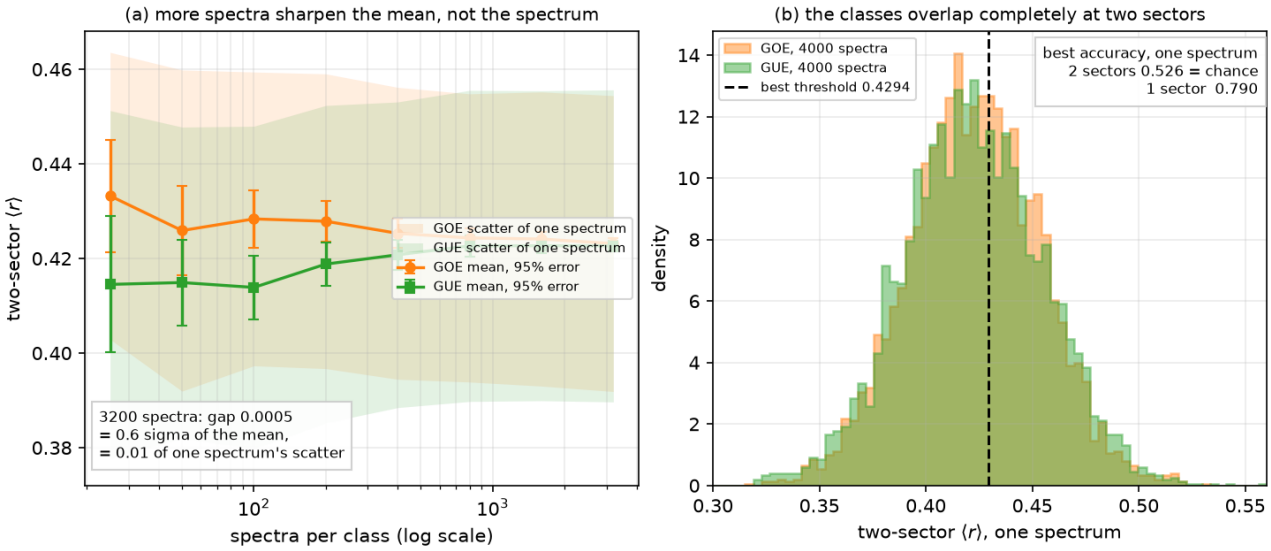


Figure 2: Resolution of the mean gap ratio at two sectors. (a) Mean r on nested samples of 25 to 3200 spectra per class, with the 95% error of the mean as error bars and the per-spectrum scatter as a shaded band; the mean converges and its error falls with the sample size, while the band does not move. (b) The two per-spectrum distributions at 4000 spectra per class, superimposed, with the best single-threshold balanced accuracy annotated (0.526 at two sectors, chance 0.500; 0.790 at one sector, for comparison).

1.5 Contribution and scope

We provide a diagnostic that:

1. verifies that the proposed involutions commute with H and with each other, and refuses the split when they do not;
2. splits the spectrum into joint sectors, and reports the sector means against the full-spectrum mean;
3. warns when a full spectrum is contaminated by an unresolved symmetry, with a measured false-positive rate of 37 of 400 for GOE and 0 of 400 for GUE single-sector spectra at d = 60 (Appendix C);
4. estimates the number of unresolved sectors as a posterior whose interval coverage is measured at 400 spectra per row: 0.99, 0.92, 0.91, 0.96, 0.98 at 1, 2, 3, 4, 6 sectors and 0.78 for a Poisson spectrum (Appendix B);
5. measures the resolution of the statistic it uses: the best single-spectrum class accuracy is 0.790 at one sector and 0.526, that is chance, at two (Section 1.4);

6. reports the limits: the mean cannot resolve the count, the count is not recoverable from adjacent-gap ratios above two sectors, the flag's rate depends on the spectrum size, the posterior degrades on sectors that are not equal and independent, two-sector GOE and GUE are indistinguishable in the mean as well as in one spectrum, and more spectra do not change that.

The phenomenon is standard quantum-chaos practice (Giraud et al. [1]). What is local is the table measured at 4000 spectra per point to a standard error of `0.0007`, the entry points, the posterior's measured coverage, the measured resolution of the single-spectrum statistic, and the recorded limits.

2 Static desymmetrization diagnostic

2.1 Verification of the involutions and joint-sector decomposition

Given a Hamiltonian `H` and a set of involutions `{U_1, ..., U_k}`:

- **Step 1: verify.** Check `[H, U_i] = 0` for every `i`, and `[U_i, U_j] = 0` for every pair. If a commutator is nonzero the split is invalid, and the diagnostic refuses to proceed rather than returning a plausible-looking decomposition.
- **Step 2: diagonalize the involution group.** The involutions generate $(\mathbb{Z}_2)^k$; diagonalize each `U_i` and group the simultaneous eigenspaces.
- **Step 3: split.** Project `H` onto each joint eigenspace and diagonalize the block.
- **Step 4: validate.** For the σ case the sectors are `spec(A + C)` and `spec(A − C)`; that identity is checked against the blocks, not assumed. For a $\mathbb{Z}_2 \times \mathbb{Z}_2$ the four sectors are checked against a direct sum built from four known blocks.

The structural reason the split matters is worth stating explicitly. An involution is an operator σ with

σ^2 `= 1.`

If the Hamiltonian commutes with it, `[H, σ] = 0`, the Hilbert space decomposes into the two eigenspaces of σ, with eigenvalues `+1` and `-1`. In the eigenbasis of σ, the Hamiltonian is block diagonal:

$$H = \begin{pmatrix} H_+ & 0 \\ 0 & H_- \end{pmatrix}$$

The full spectrum of `H` is the union of the spectra of H_+ and H_-. A spacing statistic measured on the full spectrum is therefore measured on a superposition, not on either sector. That is the silent failure mode: a value of $\langle r \rangle$ `= 0.42` looks like a legitimate measurement, and it is exactly what a two-sector superposition produces.

In the application that motivated this paper, σ is the map

`σ(s) = 1 − conj(s),`

whose fixed locus is the critical line `Re(s) = 1/2`. That involution came from an abandoned route to the Riemann hypothesis (§1.2); it is used here as a diagnostic tool, not as a proof route.

The refusal in Step 1 is not a formality. A split routine that skips it returns arrays that are not sector spectra, and the error is invisible downstream: on a fixture whose $\|[H,U]\|/\|H\|$ is `1.14 ... 1.18`, the unchecked split discards exactly the mass that makes the blocks non-sectors.

2.2 Contamination detection from a single spectrum

Two entry points cover a caller that has only a spectrum:

- `contamination_warning(H, U)`: when a candidate involution is known. It verifies the commutation, splits, and compares the full spectrum with the resolved sectors. This is the reliable half: the comparison is within one Hamiltonian and needs no calibration.
- `detect_hidden_symmetry(eigenvalues)`: when no involution is known. It computes the full-spectrum $\langle r \rangle$ and flags a value below the single-sector floor, where the floor scales with the number of levels in the spectrum (§5.1). A bare spectrum is screened under the null hypothesis that it is a single sector, so the size that enters the floor is its total length.

The flag is not a classifier (Fig. 3). `37` of `400` pure single-sector GOE spectra at `d = 60` also fall below the floor, a rate of `9.2%` with a 95% binomial interval of `6.6%` to `12.6%`, and `0` of `400` GUE spectra do (Appendix C). The rate is that high because the one-sector GOE value is `0.530` with a per-spectrum scatter of `0.039`, and the floor at 60 levels is `0.48`: a third of a distribution width below the mean is a tenth of the distribution. A low $\langle r \rangle$ is also what an integrable system produces. Spacing statistics cannot separate the two, and the warning says so in its own text.

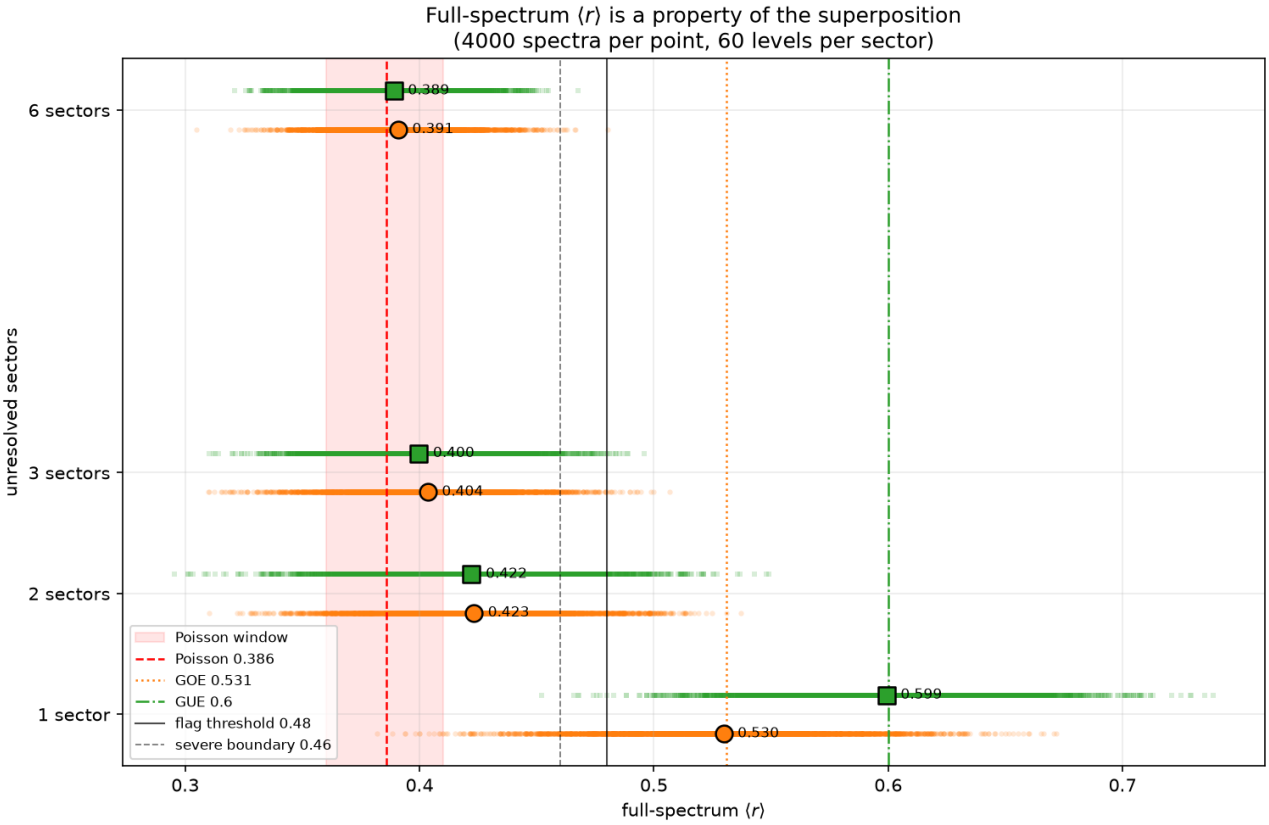


Figure 3: Distribution of the full-spectrum mean gap ratio for four sector counts and both block classes, one dot per simulated spectrum (4000 per point), with the per-point mean marked. The Poisson window is shaded, the flag threshold at 60 levels (0.48) and the severe-contamination boundary (0.46) are drawn. At two sectors the GOE and GUE clouds coincide, so a class test on the full spectrum is uninformative rather than noisy.

2.3 Bayesian estimation of the sector count

The mean $\langle r \rangle$ cannot resolve the sector count. The mean is monotone in the count at the ensemble level, but the per-spectrum scatter exceeds the gaps between successive counts, so a single spectrum's

mean does not carry its count. The sequence of ≈240 gap ratios carries more.

The reason is information. The mean collapses about 240 ratios into a single number, and the collapse is many-to-one: a three-sector and a six-sector superposition both land near 0.39, so no function of the mean can tell them apart. The sequence keeps the ratios themselves, and the way one ratio follows the next carries the sector count that the mean discards.

Model. A first-order Markov model of the sequence,

```
r_{n+1} | r_n ~ f(r_{n+1} | r_n; k),
```

where k is the number of unresolved sectors, restricted to k = 1, ..., 6, plus a Poisson hypothesis standing for "integrable, or more sectors than are distinguishable from it". Concretely, the ratios are placed on a fixed grid of bins, and for each hypothesis the conditional frequency f(r_{n+1} | r_n; k) is estimated by counting transitions between bins across 200 simulated superpositions per hypothesis. The block class is marginalised at equal weight, so the posterior answers only the question asked: how many sectors, not what class. The posterior is a plug-in estimate with a uniform prior over the seven hypotheses: the transition frequencies are fitted once and then used as if they were exact, not marginalised over, which is why its interval coverage is measured rather than assumed (§2.4).

Why first order? Modelling the ratios as independent is measurably overconfident: coverage 0.70 at a nominal 0.90, because consecutive ratios share a spacing and the independent product counts the same information twice. The first-order chain is the smallest model that keeps the dependence, and the coverage of the resulting interval is measured rather than assumed.

Output. A posterior over {1, 2, 3, 4, 5, 6, ≥6/poisson} and the shortest contiguous interval carrying 95% of its mass. The open label exists so that the estimator is *allowed* to say "not a finite superposition"; it does not claim to distinguish seven sectors from an integrable spectrum.

2.4 Measured coverage of the credible interval

At a nominal 95% interval, over 400 spectra per row (Fig. 4):

true sectors	coverage	posterior on the truth	modal interval	P(lower ≥ 3)
1	0.99	0.97	1..1 (376 of 400)	0.00
2	0.92	0.66	2..2 (146 of 400)	0.08
3	0.91	0.24	3..≥6 (138 of 400)	0.60
4	0.96	0.19	3..≥6 (166 of 400)	0.80
5	0.99	0.22	3..≥6 (166 of 400)	0.85
6	0.98	0.22	3..≥6 (174 of 400)	0.90
Poisson	0.78	0.24	3..≥6 (150 of 400)	0.94

Reading the table:

- **One sector is decisive.** p = 0.97 on the truth, interval 1..1 in 376 of 400 draws, and the posterior on 1 is below 0.001 for every other truth, Poisson included.
- **Two sectors is bounded below, not counted.** The lower end of the interval is 2 in 92% of draws, but the upper end ranges from 2 (146 of 400 draws) to the open label and beyond, so "at least two" is defensible and "exactly two" is not. The posterior on the truth is 0.66.
- **Three or more is not counted.** The posterior reports ≥3, and 3 through 6 are indistinguishable. The probability that the lower end is at least 3 rises with the truth (0.60, 0.80, 0.85, 0.90 at 3, 4, 5, 6), which is a weaker and more useful statement than a point estimate.
- **Poisson is the hard row.** The lower end is at least 3 in 94% of draws, so an integrable spectrum is described as a superposition of three or more sectors, because that is what it looks

like. Its coverage, 0.78, is the honest number for the row: in the other 0.22 of draws the interval excludes the open label, and that residual is the false-positive rate of the posterior.

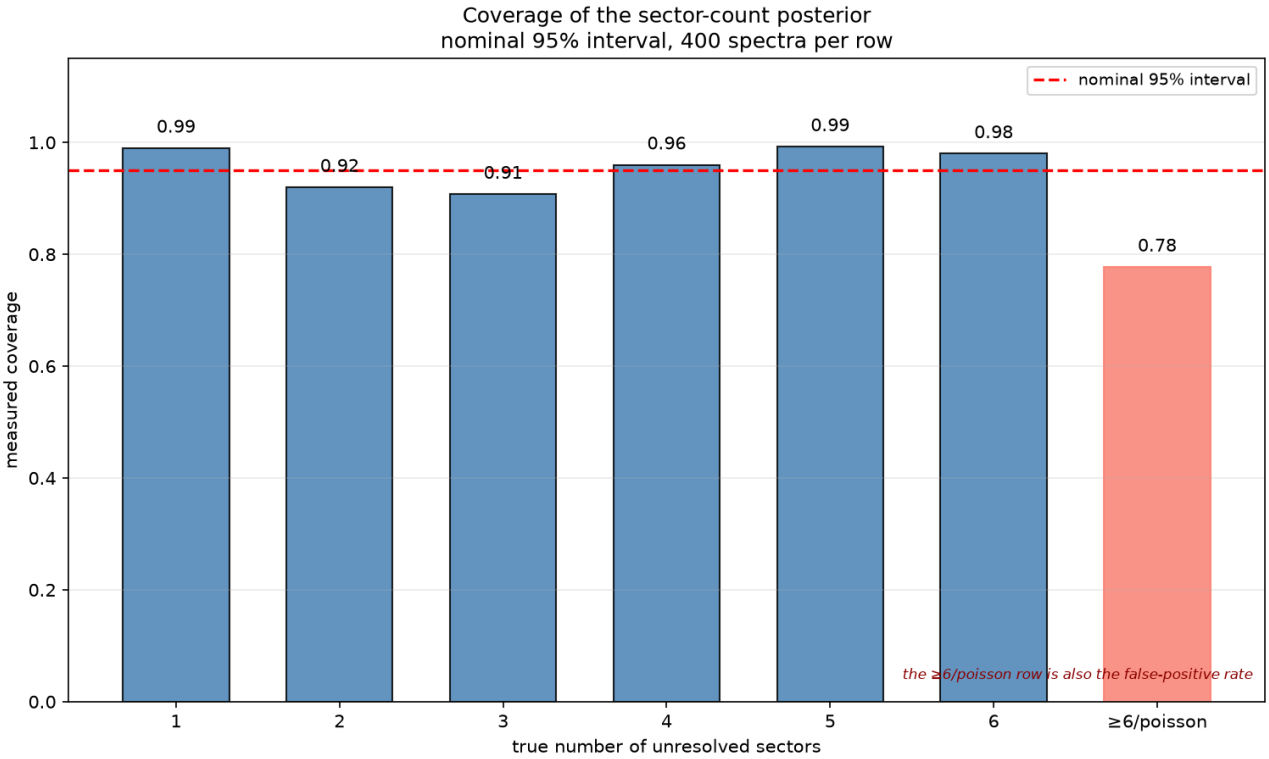


Figure 4: Measured coverage of the sector-count posterior at a nominal 95% interval, 400 spectra per row, against the nominal level (dashed). The open-ended label is the hard row: an integrable spectrum is covered by its own label in only 0.78 of draws, so in 0.22 of draws the posterior reports a finite count, and that residual is also its false-positive rate.

2.5 The count from higher-order spacing ratios

The adjacent-gap ratio is the first member of a family. With spacings `s_i = E_{i+1} − E_i`, the `k`-th order spacing ratio is

`r_i^(k) = (E_{i+2k} − E_{i+k}) / (E_{i+k} − E_i),`

folded to `(0, 1]` with min/max so that `k = 1` reproduces $\langle r \rangle$. Tekur, Bhosale and Santhanam [10] showed that the `k`-th order distribution of a composite spectrum is the nearest-neighbour distribution with a modified Dyson index, which is why the family can carry a count that `k = 1` cannot.

Model. The mean ratio at each order `k = 1, ..., 6` is one feature. For each hypothesis (1 to 6 sectors, plus Poisson) the feature vector is fitted on 200 simulated spectra with the class alternated, and the posterior is Gaussian naive Bayes over the six per-order means, with one mean and one standard deviation per hypothesis and order and a uniform prior over the seven hypotheses. It is a plug-in classifier like the one in Section 2.3, and its coverage is measured the same way, at 120 spectra per row:

With `k = 1` alone the model is the adjacent-gap limit and reproduces the negative result of Section 2.4: single-spectrum balanced accuracy is 0.66 for two versus three sectors, 0.63 for three versus six and 0.55 for four versus six. With orders one to six the same model reaches 0.81, 0.94 and 0.82. The count that the adjacent-gap ratio cannot carry above two sectors is carried by the family, and the Poisson row improves from 0.78 to 0.96: an integrable spectrum is now identified as such in 0.90 of draws, against 0.24 for the adjacent-gap model.

true sectors	coverage	posterior on the truth
1	1.00	0.99
2	0.90	0.76
3	0.78	0.47
4	0.82	0.40
5	0.87	0.31
6	0.88	0.48
Poisson	0.96	0.90

None of the higher-order machinery is new: it is Tekur, Bhosale and Santhanam's, and Bhosale's, and Section 8 sets the two recent papers that use it against this one. What is contributed here is the measured boundary of the `k = 1` statistic, and the demonstration that the family crosses it on the same spectra used throughout.

3 Symmetry dynamics under the double-bracket flow

3.1 Definition of the flow

The multi-copy imaginary-time circuits are built on the double-bracket flow

$\mathrm{d}\rho/\mathrm{d}\beta = [[\rho, H], \rho] = 2\rho H\rho - H\rho^2 - \rho^2 H.$

This flow has a property the static analysis does not: it preserves the sector structure when the state commutes with the involution.

3.2 Isospectrality and preservation of the sector populations

Measured over $\beta = 15$ with RK4 at 3000 steps:

quantity	drift
spectrum of ρ	3e-12
trace	1.3e-15
non-Hermiticity	2.8e-17
sector populations, when $[\rho, \sigma] = 0$	< 1e-8

The populations panel of Fig. 5(a) uses a state with equal mass in the two sectors that is *not* an eigenstate of `H` ($\|[\rho, H]\| = 1.46$): the flow does real work on it and the populations still do not move.

3.3 Sector selection from a non-invariant state

When the state does not commute with σ, the flow drives it into a single sector:

quantity	before	after
coherence $\|[\rho, \sigma]\|$	1.41	3e-3
populations	(0.50, 0.50)	(1.00, 0.00)

The flow selects a sector rather than preserving one. The direction is expected rather than surprising, and this section records the measurement, not a discovery: the commutant of an involution is closed under products and commutators, so a commuting state cannot leave it, and a state with both parity components flows to the ground state, which has definite parity. What is measured here is the selection itself. From an asymmetric start the flow does not break the symmetry; it settles into a definite sector.

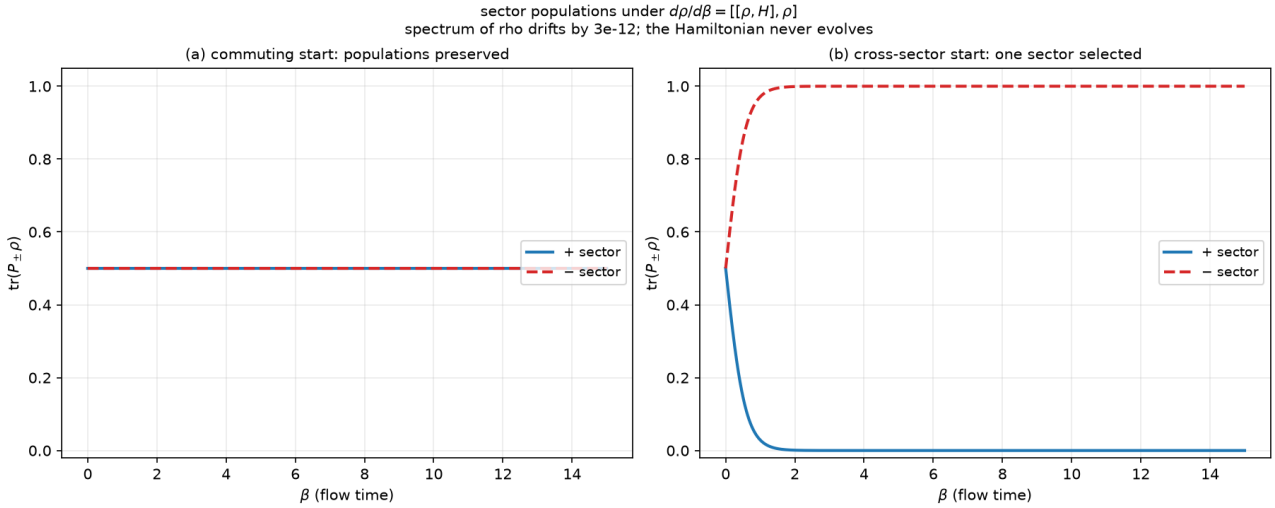


Figure 5: Sector populations under the double-bracket flow for (a) a state that commutes with the involution and (b) one that does not. Panel (a): populations (0.50, 0.50) are preserved to rounding error while the state itself evolves (the initial state is not an eigenstate of H). Panel (b): the coherence falls from 1.41 to 3e-3 and the population is driven to (1.00, 0.00). The flow is isospectral in the state and never acts on the Hamiltonian.

3.4 The flow cannot repair a contaminated spectrum

In neither case does the flow touch the Hamiltonian. `H` is an input, the state is the variable, and the flow is isospectral in the state. The class is a property of `H`, and `H` never evolves. The flow cannot repair a contaminated spectrum, and no amount of imaginary-time evolution changes the number $\langle r \rangle$ of a fixed Hamiltonian.

That is a precise statement of what the flow can and cannot do. It is the kind of limit that makes a tool trustworthy.

3.5 Machine-checked invariance of the commutant

The invariance step is machine-verified in `lean/RiemannFramework/DynamicalDesymmetrisation.lean`:

- `commute_doubleBracketIncrement`: if ρ and `H` commute with an involution, then so does the double-bracket increment;
- `commute_euler_step`: the same holds for one Euler step.

Both are `sorry`-free and depend only on `propext`, checked by `#print axioms`.

4 Non-reproduction of a quoted multi-copy identity

A description of the multi-copy imaginary-time circuits quotes an identity for the first-order term. The description has no citable source, and no published derivation of the identity was found for this paper, so it is reproduced here as quoted and tested as quoted:

$\mathrm{Tr}_2(u\ \rho_1 \otimes \rho_2\ u\dagger) = \rho_1 - \varepsilon[[\rho_1, H], \rho_1] + O(\varepsilon^{3/2})$.

The identity is tested against the two Hermitian-generator circuits ($\exp(i\sqrt{\varepsilon}(H_1 - H_2))$ and the Trotter split of the two phases) by fitting $\rho(\varepsilon) = \rho + \sqrt{\varepsilon}\ A + \varepsilon\ B + \ldots$ from three scales. In both cases:

- the first-order term is **not** zero: it is $A = i[H, \rho_1]$, which the identity does not have;

- the second-order term is **not** $\pm$[[ρ_1, H], ρ_1].

A swap variant is also fitted in the code, but $(H_1 - H_2)S$ is not Hermitian, so the numeric construction is not the exponential of that generator; it is not used as evidence for or against the identity.

The first-order term is exact, not a numerical artefact. Writing $u = \exp(i\sqrt{\varepsilon}\, G)$ and expanding to first order,

$\mathrm{Tr}_2[u\ (\rho_1 \otimes \rho_2)\ u^\dagger] = \rho_1 + i\sqrt{\varepsilon}\ \mathrm{Tr}_2[G, \rho_1 \otimes \rho_2] + O(\varepsilon)$.

For $G = H_1 - H_2$, $\mathrm{Tr}_2[H_1, \rho_1 \otimes \rho_2] = [H, \rho_1]$ (because $\mathrm{Tr}\ \rho_2 = 1$) and $\mathrm{Tr}_2[H_2, \rho_1 \otimes \rho_2] = 0$, so $A = i[H, \rho_1]$. A generator can have $A = 0$ for every ρ_1 only if $\mathrm{Tr}_2[G, I \otimes \rho_2] = 0$, which $H_1 - H_2$ does not satisfy. The quoted identity therefore does not hold for these circuits and is **deprecated**; the flow itself and its invariants (Sections 3.2–3.5) are unaffected.

The numerical checks in `dynamical_desymmetrisation.py` (`two_copy_expansion`, `first_order_term`) pin both the fitted coefficient and the analytic one, so the deprecation is a testable claim rather than a reading of the literature.

5 Limitations and scope

limit	measured	consequence
the mean cannot resolve the count	the mean is monotone but the scatter exceeds the gaps between counts	the posterior must use the sequence, not the mean
two-sector GOE/GUE indistinguishable	0.4232 vs 0.4222, 1.4 standard errors of the two means	a class test on a contaminated spectrum is uninformative
more spectra do not fix the two-sector class test	best single-spectrum balanced accuracy 0.526 at 4000 spectra per class	the diagnostic must split the spectrum, not average it
the count is not recoverable from adjacent-gap ratios above two	3...6 indistinguishable to $\langle r \rangle$ and to the ratio sequence	report $\geq$3, not a number; long-range statistics do better (§5.3)
false positive at the floor	37 of 400 GOE, 0 of 400 GUE, at d = 60	the flag is a hint, not a decision, and its rate depends on the spectrum size (§5.1)
Poisson read as a finite superposition	coverage 0.78, so a finite count in 0.22 of draws	an integrable spectrum looks contaminated about a fifth of the time
the flow cannot repair contamination	H is an input	the class is a property of H; H does not evolve
the model is a plug-in fit	200 spectra per hypothesis	the coverage is measured rather than asserted, and under-covers at two and three sectors (0.92, 0.91)

5.1 The flag's rate is a function of the spectrum size

The contamination floor is the fixed 0.48, which is the one-sector GOE mean minus 1.3 per-spectrum standard deviations at d = 60. The scatter of a mean over n ratios falls as $1/\sqrt{n}$, so the same absolute floor over-flags at small sizes and under-flags at large ones (measured, 400 spectra per point, GOE):

levels per sector	one-sector std	flag rate at fixed 0.48	two-sector superpositions missed	flag rate at a scaled floor
15	0.087	0.29	0.19	0.12
60	0.039	0.11	0.04	0.11
240	0.020	0.013	0.00	0.12
480	0.014	0.003	0.00	0.12

The fifth column keeps the same 1.3 z-score at every size (R_GOE − 1.3·σ(n) with σ(n) = 0.0393·√(60/n)), and its rate is flat at about 0.12. That is the floor the detector uses, so its threshold moves with the spectrum and the flag rate does not; the fixed 0.48 remains the reference value at 60 levels, where the two coincide.

The scaled floor is not free, and the trade runs in both directions. At small sizes it sits below 0.48, so it flags fewer pure spectra but misses contamination: at 15 levels per sector it misses 0.46 of two-sector superpositions against 0.19 for the fixed floor, and at 30 levels 0.21 against 0.12. At long sizes it sits above 0.48, so it flags more pure spectra, about 0.12 at 240 and 480 levels against 0.013 and 0.003, while missing no two-sector superposition under either floor. The two floors are mirror images: the fixed one is talkative at short sizes and quiet at long ones, the scaled one the reverse. The scaled floor is chosen for a rate that does not depend on the size, and both directions of the trade are stated here rather than only the one that flatters it. A caller should therefore report the z-score together with the missed rate at the size in hand. The calibration is in levels per sector, which is the size of one block; a caller who knows only the total length of a spectrum should pass that length, because that is what the null hypothesis, a single sector, assumes.

Appendix C's 9.2% and the 0.11 in this table measure the same quantity at different sample sizes: their difference is sampling noise, with a standard error of about 0.015, not a measurement that moved.

5.2 Robustness of the posterior to its own simulator

The posterior is trained on equal-sized, independent, class-balanced sectors at 240 levels. Real symmetries give unequal sectors, mixed classes and awkward densities, so the same posterior is run over the symmetric baseline and three departures (400 spectra each; the baseline uses GOE draws, so its mass on the truth differs from the class-marginalised coverage of Section 2.4):

scenario	truth	posterior on the truth	coverage
equal sectors	3	0.30	0.90
unequal sectors (120/80/40)	3	0.26	0.90
mixed class (GOE + GUE)	2	0.69	0.95
awkward density (one sector compressed)	3	0.26	0.92

The departures degrade the three-sector rows modestly and never improve on the baseline; the two-sector mixed-class row stays the strong case. The coverage numbers in Sections 2.4 and 1.5 are therefore a property of the symmetric simulator, not a promise about a real spectrum, and Appendix D's device descriptions are stated as an outlook for exactly that reason.

A separate axis is symmetry breaking in the data themselves. Writing H(g) = H0 + g·Δ·V, with H0 two 120-level GOE blocks, V a full GOE matrix and Δ the mean level spacing of H0, the dimensionless g is the mixing matrix element in units of the level spacing. Measured over 60 draws per point with the exact-sector posterior (docs/symmetry_breaking.md): the MAP accuracy is 0.62 at g = 0, rises to 0.87 at g = 0.2, and falls to 0.53 at g = 0.3 and 0.18 at g = 0.5; the 95 % coverage falls from 0.88 to 0.70 at g = 0.3. The critical value under the stated thresholds is g_c = 0.3, i.e. the symmetry must be intact to better than about a third of a level spacing before the posterior's interval is trustworthy. The small improvement below g = 0.2 is measured and reported rather than smoothed away; it is a finite-size training effect, not evidence that noise helps.

5.3 The limit is the ratio, not the spectrum

"Not recoverable above two" is a property of the adjacent-gap ratio, not of the spectrum. The ratio is a short-range statistic, and a three-or-more-sector superposition and a Poisson spectrum have the same adjacent gaps. Long-range statistics do not (measured, window length 20, 200 spectra per point):

spectrum	$\langle r \rangle$	number variance $\Sigma^2(20)$	Dyson–Mehta $\Delta_3(20)$
1 sector	0.530	0.98	0.22
2 sectors	0.423	1.65	0.39
3 sectors	0.403	2.24	0.51
6 sectors	0.392	3.76	0.74
Poisson	0.386	12.41	1.23

$\Sigma^2(20)$ reads a Poisson spectrum as 12.4 and a six-sector union as 3.8, where the ratio reads both as 0.39. Long-range statistics therefore separate Poisson from a union of sectors almost perfectly, which is the comparison the ratio fails and the one a complete instrument should add. The posterior is the instrument when the symmetry is unknown and only the gap ratios are available; the number variance and the Dyson–Mehta statistic are the instruments when a bare spectrum must be read as Poisson or as a union of rigid sectors. A complete diagnostic reports both, because the two statistics fail in different places. The posterior in this paper uses the ratio sequence alone, so its "not counted above two" is a statement about that statistic and nothing stronger.

The number variance is also sensitive to the unfolding. With the polynomial unfolding used here the Poisson value is 12.2 at a window of 20, against 17.8 with no polynomial unfolding and 20 for a pure Poisson process, so the absolute values and the factor of three depend on the unfolding while the ordering does not. Table 5.3 compares means; one spectrum at a time, with the balanced accuracy of Section 1.4, the two statistics separate three sectors from six at 0.90 (Σ^2) and 0.93 (Δ_3), and six sectors from Poisson at 0.98 and 0.96.

6 Comparison with explicit desymmetrization

The standard alternative is explicit desymmetrization: find the symmetry, block-diagonalize, analyse the blocks.

method	cost	when it applies
explicit	one block-diagonalization	the symmetry is known
implicit (this diagnostic)	one eigenvalue spectrum	the symmetry may be unknown

The implicit test is cheaper but coarser. It flags contamination when the symmetry is unknown; it does not replace explicit desymmetrization when the symmetry is known. When the symmetry is known, the explicit route also gives the class of each sector, which the implicit route cannot recover at two sectors.

The rule is the title of the document: **desymmetrize before testing the class.**

7 Reproducibility

7.1 Software and commands

Every number, table and figure in this paper is produced by the *Riemann Framework* repository, released under the GNU General Public License v3.0 and available at

https://github.com/mixic/riemann-framework

The repository is a Python package (`python/riemann_framework`) with a command-line script layer (`scripts/`) and a test suite (`python/tests/`). Three modules carry the results of this paper:

Four commands regenerate the paper's content. They are run from the repository root and write into `output/`:

module	what it provides
`python/riemann_framework/statistics.py`	the gap ratio `r_ratio` and its mean, computed without unfolding
`python/riemann_framework/desymmetrisation.py`	the split (`desymmetrise`, `sector_blocks`), the two warnings, the calibration table and the sector-count posterior (`sector_count_posterior`)
`python/riemann_framework/dynamical_desymmetrisation.py`	the double-bracket flow, its invariants, and the multi-copy circuit check of Section 4

command	what it produces
`python scripts/run_desymmetrisation_demo.py`	Fig. 1 and the 4000-seed calibration table, printed
`python scripts/generate_desymmetrisation_figures.py`	Figs. 2 to 4 and the coverage table, printed
`python scripts/run_dynamical_desymmetrisation_demo.py`	the flow trajectory and the circuit-expansion table (a working demo, not a manuscript figure)
`python scripts/build_manuscript_pdf.py`	this paper as a PDF

The figures are not drawn from stored numbers. Every point is computed when the script runs, so the scripts and the tests, rather than the image files, are the record of the measurement; a figure that could not be regenerated would be an assertion, not evidence.

The test suite is run from the `python/` directory, and each file pins one part of the paper:

```
cd python
python -m pytest tests/test_desymmetrisation.py -v            # split, table, posterior
python -m pytest tests/test_dynamical_desymmetrisation.py -v  # the flow
python -m pytest tests/test_desymmetrisation_figures.py -v    # the figures
python -m pytest tests/ -q                                    # the whole repository
```

The PDF itself is built by `scripts/build_manuscript_pdf.py` and checked by `scripts/verify_pdf.py`, which reads the built file back and asserts its page count, its figure placement, its text layer and the absence of build artefacts. `scripts/build_and_verify.py` runs the two in sequence and stops at the first failure.

7.2 Computational environment

Measured on the machine that produced the tables:

component	version
Python	`3.14.7`
NumPy	`2.5.3`
matplotlib	`3.11.2` (figures only)
pytest	`9.1.1`
mpmath	`1.3.0` (not used by these modules; used elsewhere in the repository)

SciPy is installed in the development environment but is **not** used by the desymmetrisation or dynamical modules; no result here depends on it.

7.3 Data and figure files

The measured table, the coverage table and the false-positive rate are produced by `scripts/generate_desymmetrisation_figures.py` and pinned by `tests/test_desymmetrisation_figures.py`. The resolu-

tion figure is produced by `scripts/generate_seed_study_figure.py` and pinned by the same test file. Every number in the figures is computed at draw time; none is typed into the plotting code.

figure	file	contents
Fig. 1	`output/desymmetrisation.png`	$\langle r \rangle$ against unresolved sectors, 4000 seeds per point
Fig. 2	`output/desymmetrisation_seeds.png`	nested samples, and the two-sector overlap
Fig. 3	`output/desymmetrisation_warning.png`	one dot per spectrum, four sector counts, both classes, Poisson window, both thresholds
Fig. 4	`output/desymmetrisation_coverage.png`	measured coverage against the nominal 95% level
Fig. 5	`output/desymmetrisation_flow.png`	sector populations under the flow, commuting and cross-sector starts

These five files, the scripts that produce them and the tests that pin them are all versioned in the repository named in Section 7.1, at `https://github.com/mixic/riemann-framework`, together with the Lean development referred to in Section 3.5. No external data set is used: the spectra are generated from the seed sequence fixed in the scripts, so a fresh checkout reproduces the tables and the figures without further input.

8 Relation to prior work

The phenomenon this paper addresses is not new. The superposed spectra of independent symmetry sectors were studied by Rosenzweig and Porter [3] and entered quantum chaos through the Bohigas–Giannoni–Schmit conjecture [4]; the gap-ratio statistic used throughout is from Atas et al. [5] and Oganesyan and Huse [6]. The specific observation that a superposed spectrum biases the gap-ratio distribution away from the single-sector class is due to Giraud et al. [1]. This paper takes that phenomenon as given and measures its consequences.

Two recent works address the same structural fact, that discrete symmetries split the Hilbert space into invariant sectors, and that a composite spectrum does not preserve the spectral correlations of its blocks, and cross the boundary this paper measures, using higher-order statistics rather than the adjacent-gap ratio.

The higher-order family itself is older, and the two papers below build on it. Tekur, Bhosale and Santhanam [10] introduced the `k`-th order spacing ratio and derived the modified Dyson index for the Gaussian and circular ensembles (2018). Tekur and Santhanam [9] then showed that a spectrum composed of `k` independent level sequences of equal level density has a `k`-th order spacing-ratio distribution equal to its nearest-neighbour counterpart with that modified index, which is how the symmetry structure of a measured spectrum can be read without desymmetrization (2020). Bhosale [11] carried the construction to superposed spectra (2021). The statistics the two papers below use, and those measured in Section 2.5, are theirs.

Vyas, Tekur and Santhanam (arXiv:2610.03108) use higher-order spacing ratios (HOSR) and a Best Subset Technique (BST). For order `k` they define

`r_i^(k) = (E_{i+2k} − E_{i+k}) / (E_{i+k} − E_i),`

and fit the empirical `k`-th order spacing-ratio distribution to the Wigner surmise with an effective Dyson index β', minimising the Kullback–Leibler divergence

`D_KL(k, f) = Σ_r P_emp^(k)(r; f) · log[ P_emp^(k)(r; f) / P(r; β′) ]`

over a fraction `f` of the spectrum that selects the maximally overlapping part of the blocks. Their central self-consistency relation is

`P_k(r, β = 1, m) = P(r, β′ = k = m) (GOE)`
`P_k(r, β = 2, m) = P(r, β′ = k+1 = m+1) (GUE)`

so that for `k = m` in the GOE case (or `k+1 = m+1` in the GUE case) the number of sectors `m` becomes directly readable. The `k`-th order ratio sees the sector count where the adjacent-gap ratio does not.

Rout and Bhosale (arXiv:2510.00503) study higher-order spacings in superposed spectra of random matrices of the same class, with equal-sized subsectra, and test the result on the quantum kicked top. Their setting is therefore the symmetric case that the best-subset technique generalises: the unequal-dimension problem is treated by Vyas et al. through the optimal fraction `f*`, not by Rout and Bhosale.

The relation between those methods and this paper is precise and complementary, not competitive. The adjacent-gap ratio is the `k = 1` case of the higher-order family. This paper measures where the `k = 1` statistic stops carrying the sector count: at three or more sectors the $\langle r \rangle$ values (`0.404`, `0.396`, `0.391`) lie closer together than the per-spectrum scatter (`0.017` to `0.039`) that one spectrum carries, so no ensemble average can separate them.

The crossing is measured here rather than quoted. Section 2.5 extends the posterior to orders one to six, and on one spectrum the difference is direct: the mean ratio alone separates two from three sectors at `0.66` balanced accuracy, three from six at `0.63` and four from six at `0.55`, while orders one to six separate them at `0.81`, `0.94` and `0.82` (200 spectra per class). The count that the adjacent-gap ratio cannot carry above two sectors is carried by the family, and the statement "not recoverable" belongs to `k = 1`, not to the spectrum.

The involution σ plays a different role in the two settings. In this paper σ is a concrete, verifiable operator: `[H, σ] = 0` and `[U_i, U_j] = 0` are checked before the split, and the involution is the key that guarantees the block structure. In Vyas et al. the symmetry is more general and less explicit: the sectors may arise from several discrete symmetries (parity, reflection, and others), their dimensions may be unequal, and the BST recovers the count `m` without knowing the symmetry operators. The involution used here is the simplest case of the general problem those papers solve: `m = 2`, equal dimensions, one explicit $\mathbb{Z}_2$ symmetry.

The contribution of this paper is therefore a measurement of that case, not a method that generalises it. It reports the interval coverage of a Bayesian posterior over the ratio sequence (`0.99` to `0.78`, at 400 spectra per row), the separation of precision from resolution (4000 spectra determine the mean to `0.0007`, while the best single-spectrum class accuracy at two sectors is `0.526`, which is chance), and the explicit statement that the ratio stops carrying the count above two sectors, where the Dyson–Mehta number variance continues to separate Poisson from a six-sector union by a factor of three.

reference	contribution	what this adds
Giraud et al. [1]	the phenomenon: superpositions bias the ratio	the measured table, the entry points, the posterior, the coverage
Rosenzweig–Porter [3]	block superpositions in RMT	the diagnostic form, not the theory
Mehta [2]	the reference distributions	the calibration values
Atas et al. [5]	the `r`-ratio as a statistic	the inversion attempt and its measured failure
Vyas, Tekur, Santhanam [7]	sector count from higher-order ratios, unequal sectors	the measured coverage and the stated ratio-limit
Rout, Bhosale [8]	higher-order spacings, quantum kicked top	the Bayesian posterior and the resolution measurement

The phenomenon is not new. The boundary is what this paper measures.

9 Conclusion

A spacing statistic measured on a Hamiltonian that commutes with an involution is measured on a superposition. The standard $\langle r \rangle$ cannot distinguish a Poisson spectrum from a union of unresolved sectors, and the failure is silent.

We provide a diagnostic that verifies the symmetries, splits the spectrum, flags contamination, and estimates the sector count as a posterior with a measured coverage. The single-sector case is decisive; the two-sector case is bounded below but not counted; from three sectors upward the count is not recoverable from adjacent-gap ratios alone, and long-range statistics do what the ratios cannot. Two-sector GOE and GUE are indistinguishable, in one spectrum and in the mean of thousands. The false-positive rate is stated, as a function of the spectrum size. The limits are reported rather than hidden.

The measurement that decides how the tool may be used is the resolution, not the precision: 4000 spectra per point determine the mean to `0.0007`, and the best single-spectrum class test at two sectors is still `0.526`, which is chance. Averaging buys a better-known ensemble; it does not buy a classifiable spectrum. The instrument therefore resolves the symmetry, and the number it reports is a flag rather than a classifier.

The companion dynamical analysis shows that the double-bracket flow preserves sector populations when the state commutes with the involution, but selects a sector when it does not, so the flow cannot repair a contaminated spectrum, because the class is a property of the Hamiltonian and the Hamiltonian does not evolve.

The rule is simple: desymmetrize before testing the class.

The diagnostic began as an error made on a candidate operator; it ends as a tool that makes that mistake visible, measurable, and stated.

Appendix A: The full measured table

At 4000 seeds per point, `60` levels per sector, with the per-spectrum standard deviation of the mean $\langle r \rangle$:

sectors	GOE $\langle r \rangle$	GOE std	GUE $\langle r \rangle$	GUE std
1	`0.5297`	`0.0393`	`0.5993`	`0.0374`
2	`0.4232`	`0.0312`	`0.4222`	`0.0329`
3	`0.4035`	`0.0250`	`0.3997`	`0.0246`
4	`0.3964`	`0.0213`	`0.3932`	`0.0209`
6	`0.3909`	`0.0174`	`0.3890`	`0.0172`

The standard error of each mean is the standard deviation divided by the square root of `4000`, that is by `63.2`, so no entry is uncertain by more than `0.0007`.

The two-sector gap between classes (`0.0010`) is `1.4` standard errors of the two means and one thirtieth of either standard deviation, which is the quantitative form of "the class is erased". Adding spectra does not change it: Section 1.4 measures the same gap on a nested series from 25 to 3200 spectra, and it stays consistent with zero at every sample size.

Appendix B: The coverage table

At a nominal 95% interval, 400 spectra per row:

The Poisson row is the false-positive rate of the posterior: in `0.22` of draws the interval excludes the open label and reports a finite count, which is the failure mode the table exists to expose.

true sectors	coverage	posterior on the truth
1	0.99	0.97
2	0.92	0.66
3	0.91	0.24
4	0.96	0.19
5	0.99	0.22
6	0.98	0.22
Poisson	0.78	0.24

Appendix C: The false-positive rate

test	result
pure single-sector GOE spectra below the floor, d = 60	37 of 400
pure single-sector GUE spectra below the floor, d = 60	0 of 400
false-positive rate, GOE	9.2%, 95% interval 6.6% to 12.6%
false-positive rate, GUE	0%, 95% upper bound 0.75%

At 60 levels the floor is 0.48, not the reference value 0.531, because the measured one-sector GOE value at d = 60 is 0.5297 with a per-spectrum scatter of 0.0393: the floor sits 1.3 standard deviations below that mean, which is precisely why 9.2% of pure GOE spectra are flagged. A floor raised to the reference value would exchange those false positives for missed contamination, and the trade is the user's, so the rate is stated rather than tuned. The GUE side is quiet because its one-sector value is 0.5993, 3.2 standard deviations above the floor. The rate is a function of the spectrum size, not a constant of the instrument: Section 5.1 reports it across sizes, and the detector applies the scaled floor, so 0.48 is the value that scaling takes at 60 levels rather than a constant.

Appendix D: Technical applications and outlook

The diagnostic is a software method, and a method is not a device. This appendix sketches, as an outlook only, how the measured pieces assemble into a physical instrument; it makes no patent claim.

D.1 A symmetry-aware spectrum analyzer

component	technical function	what it uses here
signal acquisition	measure the energy spectrum of a sample, for instance a quantum dot array, a microwave cavity, or a material	the input spectrum
symmetry verification	test whether the spectrum commutes with a candidate involution, such as a spatial symmetry of the sample	desymmetrise(H, involutions)
sector splitting	project into joint eigenspaces once the symmetry is verified	the spec(A ± C) identity and the $Z_2 \times Z_2$ split
posterior engine	estimate the number of unresolved sectors from the gap-ratio sequence	sector_count_posterior
output	emit a contamination flag, or a posterior over the sector count	the severe / mild / none warning

The problem the instrument solves is the failure mode of this whole manuscript. An analyzer that reports one number, $\langle r \rangle$, can label a chaotic system integrable when a symmetry is unresolved. The instrument replaces the number with a verdict and a posterior whose coverage is measured rather than assumed.

D.2 A quantum-processor calibration system

A processor, for instance a superconducting qubit array, can sit in a symmetry-protected regime whose spectrum looks integrable, and control pulses calibrated against that spectrum are wrong. The device reads the spectrum, applies the diagnostic, and, on a contaminated result, drives a symmetry-breaking perturbation to resolve the sectors before re-measuring. The technical effect is control fidelity; the algorithm is the means.

D.3 A double-bracket flow controller

The flow $\mathtt{d}\rho/\mathtt{d}\beta = [[\rho, H], \rho]$ preserves or selects a sector, as measured in §3. A controller that implements the flow and monitors the coherence $\|[\rho, \sigma]\|$ can stop the evolution when the wrong sector is being selected, instead of discovering it after the prepared state is wrong.

D.4 Application domains

Three settings give the instrument its technical effect.

- **Quantum simulation and many-body systems.** A simulator that is meant to be chaotic but sits in a symmetry-protected sector can be misread as integrable; the flag and the posterior catch the misreading before it is published.
- **Microwave billiards and cavity measurements.** A cavity whose resonance spectrum is contaminated by a geometric symmetry (a mirror plane, a rotation) has a $\langle r \rangle$ that sits in the Poisson window; the diagnostic says "resolve the symmetry" instead of "integrable".
- **Quantum dots and materials spectroscopy.** A measured single-particle spectrum of a dot array or a material sample is a superposition whenever the sample carries a spatial symmetry; the posterior bounds how many sectors are unresolved.

In each case the output is not a new theory. It is a control on an old one: a number that would otherwise have been trusted is replaced by a number together with its provenance (a resolved sector or an unresolved superposition) and its uncertainty.

D.5 Scope of the claims

What the manuscript does **not** claim:

- the algorithm itself (a plug-in posterior over sector counts);
- a numerical simulation with no physical input or output;
- the mathematics of Sections 2 through 4.

A granted patent claims a physical embodiment and a technical effect, not these. This manuscript makes no patent claim and no assertion about what any office would grant; the device descriptions above are an outlook, and the quantum-processor and dot-array cases carry no experimental results behind them.

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Disclaimer

This manuscript describes a diagnostic instrument, not a result about the Riemann hypothesis. The phenomenon it addresses, that unresolved symmetries bias spacing statistics toward Poisson, is standard quantum-chaos practice. What is local is the measured table, the entry points, the posterior's measured coverage, and the recorded limits.

The diagnostic detects contamination. It does not count sectors reliably above two from adjacent-gap ratios alone, and it does not distinguish GOE from GUE at two sectors. Those limits are stated, not hidden.

The rule is: desymmetrize before testing the class.